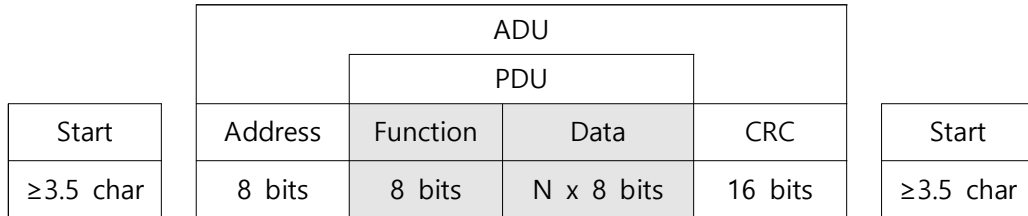


초음파 용착기 제어 프로토콜 (V2.0/250705)

- 통신방식 : Modbus 485(초기값 : RS485 / 19200 bps / 8E1) 또는 Modbus TCP

- Modbus 485 프로토콜 기본 형식



- Modbus TCP 프로토콜 기본 형식

ADU					
MBAP Header				PDU	
Transaction ID	Protocol ID	Length	Unit ID	Function	Data
16 bits	16 bits	16 bits	8 bits	8 bits	N x 8 bits

1. Address 및 Unit ID

. Modbus 485 : 1~247

. Modbus TCP : 1 (1로 고정)

2. Transaction ID

. Modbus TCP : 마스터에서 통신시작시 0에서부터 증가하며 전송. 슬레이브에서 받은 값을 사용하여 응답하여 전송값에 대한 응답을 확인한다.

3. Protocol ID

. Modbus TCP : 0 (고정값)

3. Length

. Modbus TCP : Unit ID ~ Data끝까지의 바이트 수

4. function code

Function description	Code
Read holding registers	0x03
Write single holding register	0x06

. 03(0x03) Read Holding Registers

-Request

Function code	1 Byte	0x03
Starting address	2 Bytes	0x0000 to 0xFFFF
Quantity of Register	2 Bytes	1 to 125 (0x7D)

-Response

Function code	1 Byte	0x03
Starting address	1 Bytes	2 x N
Quantity of Register	N x 2 Bytes	

. 06(0x06) Write single Holding Register

-Request

Function code	1 Byte	0x06
Register address	2 Bytes	0x0000 to 0xFFFF
Register value	2 Bytes	0x0000 to 0xFFFF

-Response

Function code	1 Byte	0x06
Register address	1 Bytes	0x0000 to 0xFFFF
Register value	N x 2 Bytes	0x0000 to 0xFFFF

3. Register address

Register address	R/W	Description	
0x00	R/W	Work count High	0 ~ 9999999
0x01	R/W	Work count Low	
0x02	R/W	out Power(W) / 0~9999	
0x03	R/W	out Amp.(A) / 0~9999 / 실제값 x 100 (123 => 1.23)	
0x04	R/W	out Freq.(KHz) / 0~9999 / 실제값 x 10 (123 => 12.3)	
0x05	R/W	out Energy(J) / 0~9999	
0x06	R/W	S/M/H : A.V.(%) / 50 ~ 100	
0x07	R/W	H : Run time(S) / 0 ~ 2000 / 실제값 x 100 (123 => 1.23)	
0x08	R/W	S/M/H : Energy(J) / 0 ~ 65535	
0x09	R/W	S/M/H : Limit time(S) / 0~10	
0x0A	R/W	S (D): Delay(D1) / 0~500 / 실제값x100 (123=>1.23)	
0x0B	R/W	S (D): Weld(D2) / 0~500 / 실제값x100 (123=>1.23)	
0x0C	R/W	S (D): Hold(D3) / 0~2000 / 실제값x100 (123=>1.23)	

0x0D	R/W	S (T): Weld(T2) / 0~500 / 실제값x100 (123=>1.23)
0x0E	R/W	S (T): Hold(T3) / 0~2000 / 실제값x100 (123=>1.23)
0x0F	R/W	S/M/H : Multi A.V. - T1(S) /0~2000/실제값x100 (123=>1.23)
0x10	R/W	S/M/H : Multi A.V. - T2(S) /0~2000/실제값x100 (123=>1.23)
0x11	R/W	S/M/H : Multi A.V. - A.V.1(%) / 50 ~ 100
0x12	R/W	S/M/H : Multi A.V. - A.V.2(%) / 50 ~ 100
0x13	R/W	Run mode (standard) (0=Delay / 1=Trigger)
0x14	R/W	Enable Energy control (0=Disable / 1 = Enable)
0x15	R/W	Enable Multi-out mode (0=Disable / 1 = Enable)
0x16	R/W	Enable Safe mode (0=Disable / 1 = Enable)
0x17	R	Model Frequency (0=15K/1=20K/2=30K/3=35K/4=40K/5=50K)
0x18	R	Model Type (0=Hand / 1=Multi / 2=Standard)
0x19	W	Reset (0=OFF / 1=ON)
0x1A	W	Seek (0=OFF / 1=ON)
0x1B	W	RUN (0=OFF / 1=ON)
0x1C	W	STOP (0=OFF / 1=ON)
0x1D	R	STATUS 0x01 : Ultra sonic ON 0x02 : E-STOP 0x04 : Overload 0x08 : Over time

4. Example

-> Work count 읽어오기

(Modbus 485)

Request		
Field	Hex	Description
Slave addr.	01	
Function	03	
Start addr. Hi	00	Reg. 00
Start addr. Lo	00	
No. of Reg. Hi	00	
No. of Reg. Lo	02	2
CRC Low	C4	
CRC High	0B	

Response		
Field	Hex	Description
Slave addr.	01	
Function	03	
Byte Count	0C	
Reg. value Hi		0 (Count high)
Reg. value Lo		
Reg. value Hi		1 (Count Low)
Reg. value Lo		
CRC Low		
CRC High		

(Modbus TCP)

Request		
Field	Hex	Description
Transaction ID	0000	
Protocol ID	0000	
Length	06	
Unit ID	01	
Function	03	
Start addr. Hi	00	Reg. 00
Start addr. Lo	00	
No. of Reg. Hi	00	
No. of Reg. Lo	02	2

Response		
Field	Hex	Description
Transaction ID	0000	
Protocol ID	0000	
Length	06	
Unit ID	01	
Function	03	
Reg. value Hi		0 (Count high)
Reg. value Lo		
Reg. value Hi		1 (Count Low)
Reg. value Lo		

-> 초음파 동작개시값 전송
(Modbus 485)

Request		
Field	Hex	Description
Slave addr.	01	
Function	06	
Start addr. Hi	00	RUN
Start addr. Lo	1B	
Reg. value Hi	00	
Reg. value Lo	01	ON
CRC Low	38	
CRC High	0D	

Response		
Field	Hex	Description
Slave addr.	01	
Function	06	
Start addr. Hi	00	RUN
Start addr. Lo	1B	
Reg. value Hi	00	
Reg. value Lo	01	ON
CRC Low	38	
CRC High	0D	

(Modbus TCP)

Request		
Field	Hex	Description
Transaction ID	0001	
Protocol ID	0000	
Length	06	
Unit ID	01	
Function	06	
Start addr. Hi	00	RUN
Start addr. Lo	1B	
No. of Reg. Hi	00	
No. of Reg. Lo	01	ON

Response		
Field	Hex	Description
Transaction ID	0001	
Protocol ID	0000	
Length	06	
Unit ID	01	
Function	06	
Reg. value Hi	00	RUN
Reg. value Lo	1B	
Reg. value Hi	00	
Reg. value Lo	01	ON

케이블 연결

신호명		색깔	Pin No	구분
Start	입력	청색	19	
		백색	23	GND
Reset	입력	녹색	15	
		주황	22	GND
Seek	입력	노랑	14	
		보라	21	GND
Overload	NPN오픈콜렉터 출력	흑색	12	
		적색	11	GND
U.S. Output	NPN오픈콜렉터 출력	회색	8	
		갈색	7	GND
RS485-COM		핑크	24	TRX+
		하늘색	10	TRX-

Modbus protocol: [Modbus_over_serial_line_V1_02.pdf](#) (2006)

Modbus protocol specification: [Modbus_Application_Protocol_V1_1b3.pdf](#) (2012).